

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTETC303 - DIGITAL ELECTRONICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 The max-term representation of a logic function is also known as:
 - A. SOP
 - B. POS
 - C. XOR
 - D. XNOR
- 2 A full subtractor requires how many inputs?
 - A. 2
 - B. 3
 - C. 1
 - D. 4
- 3 A digital comparator compares:
 - A. Voltage levels
 - B. Binary numbers
 - C. Decimal values
 - D. Memory addresses
- 4 A finite state machine with outputs depending on current state and input is:
 - A. Moore machine
 - B. Mealy machine
 - C. Ring counter
 - D. Sequence detector
- 5 Which flip-flop configuration can be used to design a shift register?
 - A. JK
 - B. T
 - C. D
 - D. SR

- 6 A ring counter with 4 flip-flops has how many states?
 A. 4
 B. 8
 C. 16
 D. Infinite
- 7 Which PLD has both programmable AND and OR planes?
 A. PAL
 B. PLA
 C. PROM
 D. FPGA
- 8 The number of states in a 3-bit counter is:
 A. 3
 B. 6
 C. 8
 D. 4
- 9 In a synchronous counter, all flip-flops are triggered by:
 A. Different clocks
 B. External pulses
 C. Same clock
 D. No clock
- 10 Which term refers to variations in the arrival time of the clock signal?
 A. Clock pulse
 B. Clock jitter
 C. Clock delay
 D. Clock edge
- 11 Which logic family has the least power consumption?
 A. TTL
 B. CMOS
 C. ECL
 D. DCTL
- 12 Which of the following is a valid VHDL data object?
 A. Port
 B. Process
 C. Signal
 D. Architecture
- Q2. Solve the following.
- A) Minimize the following Boolean function using Karnaugh map and implement the logic: $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 6, 7, 8, 9, 10, 14) + d(3, 11, 15)$ 6
- B) Explain VHDL data types and objects with examples. 6
- Q3. Solve the following.
- A) Explain the working of a BCD to 7-segment display with circuit diagram. 6
- B) Convert SR flip-flop to D flip-flop. 6
- Q4. Solve Any Two of the following.
- A) Compare TTL, CMOS, and ECL logic families 6
- B) Design a sequence detector for detecting 1011 using Mealy machine. 6
- C) Write a VHDL program for 3 to 8 Decoder. 6

Q5. Solve Any Two of the following.

- A) Draw and explain CMOS inverter and its working. 6
- B) Differentiate between PROM, PLA, and PAL with diagrams. 6
- C) Explain architecture and application of CPLD. 6

Q6. Solve Any Two of the following.

- A) Design a finite state machine to detect three consecutive 1's in a serial bit stream. 6
- B) Design a 4-bit SISO shift register using D flip-flops. Draw block and timing diagram. 6
- C) Write short notes on: 6
 - a) Tri-state logic
 - b) Wired AND and Open Collector output in TTL

*** End ***